

PANJAB UNIVERSITY, CHANDIGARH-160014 (INDIA)

(Estd. under the Panjab University Act VII of 1947 –enacted by the Govt. of India)

SYLLABI

Course Outline for
B.Voc (HARDWARE AND NETWORKING)
as per NEP 2020

w.e.f session 2024-25

Semester-I

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
BHN-101	Core Discipline	Basics of Computer System	4	--	4	80	20	--	--	100
BHN-102	Core Discipline	Fundamental of PC Hardware	2	--	4	40	10	--	--	100
BHN-102P		Practical Based on BHN- 102	--	2		--	--	40	10	
BHN-103	Core Discipline	PC Configuration	--	4	4	--	--	80	20	100
AEC / MIL	Ability Enhancement Course	English			2					50
B.Voc. MDSC 101	Multidisciplinary Course	Fundamentals of Information Technology[1]	3	-	3	60	15	-	-	75
SEC I	Skill Enhancement Course	SEC Options[2]			3					75
VAC I	Value-Added Course	VAC Options[3]			2					50
PW I	Project Work	Project Work-I & Viva- Voce	-	2	2	-	-	50	-	50
Total Credits					**24**					**600**

Semester-II

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
BHN-201	Core Discipline	Data Communication and Networks	4	--	4	80	20	--	--	100
BHN-202	Core Discipline	Computing Software	2	--	4	40	10	--	--	100
BHN-202P		Practical Based on BHN- 202	--	2		--	--	40	10	
BHN-203	Core Discipline	Network Configuration	--	4	4	--	--	80	20	100
AEC / MIL	Ability Enhancement Course	English			2					50
B.Voc. MDSC 201	Multidisciplinary Course	Soft Skills and Personality Development[1]	3	-	3	60	15	-	-	75
SEC II	Skill Enhancement Course	SEC Options[2]			3					75
SIT 201*	Internship	Summer Industrial Training (SIT)*	--	4	4	--	--	80	20	100
Total Credits					**24**					**600**

Job Role - Field Technician: Computing & Peripherals

*Summer Industrial Training of 2-4weeks in a relevant Industry after 2nd Semester Examinations during summer break. Training report by the student to be submitted within one week of start of 3rd Semester. Viva-Voce examination to be held within 3-weeks of the start of 3rd semester.

** Minimum Credits required to get Certificate Course in (HARDWARE AND NETWORKING) =48 + 4 #.
#Exit 1: Certificate Course-Student can exit after completion of 1st year having secured 48 Credits and will be awarded a UG Certificate provided they complete a vocational course/internship of 4 credits during the summer vacation of the first year, in addition to the 6 credits from skill enhancement courses earned during first and second semester.

***One credit of theory lecture is of 1 hour duration and one credit of practical is of 2 hours duration.

1. Multidisciplinary Courses (MDSC): Annexure I
2. Skill Enhancement Courses (SEC): Annexure II
3. Value Added Courses (VAC): Annexure III

ESSENTIAL COMPONENTS OF COURSE OF
B.Voc. (HARDWARE AND NETWORKING)
(SEMESTERS I & II)

PROGRAMME OUTCOMES

PO1. The primary objective of this course is to prepare students for careers as Field Technicians (Computing and Peripherals) and Hardware and Network Engineers/Administrators. These roles are crucial in the IT industry, particularly in servicing firms such as telecom companies and equipment manufacturing firms. Students will gain the necessary skills and knowledge to excel in these positions.

PO2. The course places a strong emphasis on practical, hands-on training. It recognizes the importance of providing students with an early platform to manage and troubleshoot complex failures in computer hardware and networks. By engaging in practical exercises, students will develop the skills required to analyze problems and implement effective solutions, fostering a deeper understanding of the field.

PO3. A key focus of the curriculum is to cultivate the ability of students to identify and resolve issues related to computer hardware and networks. The course structure is designed to equip students with problem- solving skills, enabling them to troubleshoot and address challenges in a real-world context. This emphasis on practical problem-solving enhances the employability of graduates.

PO4. The curriculum is well-rounded, encompassing both general education and skill development components. Beyond technical expertise, students will receive a broad education that includes relevant industry knowledge and soft skills. This comprehensive approach ensures that graduates are not only technically proficient but also well-rounded professionals capable of adapting to diverse work environments.

PO5. To bridge the gap between theoretical knowledge and practical application, the course integrates extensive practical and on-the-job training components. Students will have the opportunity to apply their theoretical learning in real-world scenarios, preparing them for the challenges they may encounter in their future roles. This hands-on experience is invaluable for building confidence and competence.

PO6. To further connect students with the professional world, the course incorporates industry interactions. Students will have the chance to engage with professionals in the field, participate in workshops, and attend seminars. This exposure facilitates networking opportunities, allowing students to learn from industry experts, stay updated on industry trends, and potentially secure internships or job placements. This connection to the industry is instrumental in preparing students for a successful transition from education to the workforce.

Semester I

	COURSE TITLE	BASICS OF COMPUTER SYSTEM
1.	**Course Code**	BHN-101
2.	**Typology of Course**	Core Discipline
3.	**Course Duration**	60 hours of lectures for Theory Theory credits =4 (1 credit= 15 hours of theory learning based on 4hrs./week) (1 Semester=15 weeks)
4.	**Credits**	4 Credits
5.	**Max. Marks and Time**	100 Marks (80TH+20IA) i) The exam of this course will be of 3hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
6.	**Job Role**	Field Technician- Computing & Peripherals
7.	**Course Objectives**	The course aims to equip students with a profound understanding of computer fundamentals, hardware, software, and networking. To empowers the ability to analyze advantages and disadvantages, design computer system architectures, apply input and output devices, and comprehend emerging technologies and well-rounded expertise in basics concepts.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1.Demonstrate a comprehensive understanding of computer fundamentals, including the historical development and characteristics of computers. CO2.Apply knowledge of input and output devices, computer memory, and different types of software in practical scenarios. CO3.Analyze and evaluate the functions and types of operating systems, data processing systems, and translators. CO4.Demonstrate proficiency in understanding computer networks, the Internet, and emerging technologies, with the ability to classify and describe different types of networks.
9.	**Pedagogy of the Course Work**	Classroom Lecture, Interactive sessions, PPT, Group Discussions etc.
10.	**Instructions for paper setters and candidates**	i) The syllabus of this paper has been divided into FOUR units. ii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. iii) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. iv) All questions carry equal marks.

11. Course Content

UNIT-I

Computers: Characteristics, Historical development of computers, Evolution of computing devices, Generations of Computers: First Generation (Vacuum Tubes, Batch Processing), Second Generation (Transistors), Third Generation (Integrated Circuits, Time-Sharing Systems), Fourth Generation (Microprocessors, Personal computers), Fifth Generation (Artificial intelligence). Hardware, Software and Firmware, Data and Information. Computer Codes: Decimal System, Binary System, Hexadecimal System, Octal System and ASCII Code. Decimal to Binary conversion, Binary to Decimal Conversion.

UNIT-II

Introduction to Computer Networks: Definition and Importance of Computer Networks, Basic Components of a Computer Network, Classification of Networks, Network Topologies. Internet: Historical Overview of The Internet, Evolution of The World Wide Web (WWW), WWW Components, Internet Governance and Standards, Internet Services and Applications (Email Protocols, Telnet, FTP, IRC, Usenet News). Internet Access Methods (Dial-Up, DSL, Cable, Fiber-Optic), Browser Technologies and Features.

UNIT-III

Computer Virus: Types of Viruses, Impact of Virus, Virus Detection and Prevention techniques, Malware. Antivirus and Its Types. Multimedia: Hardware and Software for Multimedia, Multimedia: Component of Multimedia, Basic File Formats, Audio File Formats, Image File Formats, Video File Formats. Multimedia Storage Devices.

UNIT-IV

Emerging Technologies: Basics of Internet of Things (IoT), Components of IoT, Applications of IoT, Basics of Parallel Computing and Distributed Computing, Difference between Parallel computing and Distributed Computing, Advantages and Limitations of Parallel computing and Distributed Computing, Cloud Computing: Component of Cloud Computing, Types of Cloud Computing, and their applications.

12 Suggested Resources for Reading

1. Fundamentals of Computers, V. Rajaraman, PHI Publication.
2. Computer Fundamentals, P. K. Sinha, BPB Publication.
3. Computers Today, Basandra, S. K., Galgotia Publications.
4. Fundamentals of Information Technology, Anshuman Sharma, Lakhanpal Publications.
5. PC Hardware: The Complete Reference, Craig Zacker., TMH Publication.

Note: Latest edition of text books may be used.

	COURSE TITLE	FUNDAMENTAL OF PC HARDWARE
1.	**Course Code**	BHN-102
2.	**Typology of Course**	Core Discipline
3.	**Course Duration**	30 hours of lectures for Theory + 60 hours of practicals Theory credits =2 and Practical credits=2 (1 credit = 15 hours of theory learning based on 2 hrs./week) + (1 credit = 30 hours of practical based on 4 hrs./week) (1 Semester = 15 weeks) 2-groups will be created for the purpose of Practical for each unit
4.	**Credits**	4 Credits
5.	**Max. Marks and Time**	100Marks ((40TH+10IA) +(40PR+10IA)) i) The exam of this course will be of 3hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
6.	**Job Role**	Field Technician- Computing & Peripherals
7.	**Course Objectives**	This course aims to provide comprehensive knowledge of computer peripherals, hardware components, and system management. Hardware Fundamentals enable to learn about hardware technologies, their implementation/installation, uses, troubleshooting and maintenance.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to CO1.Demonstrate a comprehensive understanding of computer hardware components, peripherals, and their functionalities. CO2.Enable the students to learn about hardware technologies and their implementation/installation, uses, troubleshooting and maintenance. CO3.Install, update, and troubleshoot drivers for various hardware components, including network adapters, sound cards, and graphics cards. CO4.Configure system settings, understand the role of BIOS, and troubleshoot common BIOS-related issues.
9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions etc.
10.	**Instructions for paper setters and candidates**	i) The syllabus of this paper has been divided into FOUR units. ii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. iii) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. iv) All questions carry equal marks.

11. Course Content

UNIT-I

Peripheral Devices: Introduction to Peripheral Devices, Functionality, Use and Types of Various Peripheral Devices (Mouse, Keyboard, Printer, and Monitor). Add-On Cards and Its Functionality: Network Interface Card, Sound Card and Graphics Card. Uninterruptible Power Supply (UPS): Features, Types (Offline & Online UPS), Advantages & Limitations of UPS. Mother Board: Definition, Importance, Basic Components, Layout of Motherboard, Types of Motherboards (AT, ATX, and BTX).

UNIT-II

Ports: Definition, Role of Ports in Computer Systems, Functions, Types of Ports: Serial Ports and its Types, Parallel Ports, and its Types. Hard Disk: Definition, Importance of Hard Disk, Structure of Hard Disk (Head, Platter, Spindle, Actuator, Tracks, Sectors), Types of Hard Disk (PATA, SATA, SCSI). RAM: Definition, Uses, Types of RAM (SRAM, DRAM, SDRAM). Power Supply Unit: Functions, Components, Power Connectors, Power Requirements for various Computer Components. Processor: Definition, Architecture, Functions, Clock Speed, Cores, and Cache Memory.

UNIT-III

Cables: Importance of Cables in Computer Systems, Types (VGA, DVI, RJ11, RJ45, HDMI, Serial, Parallel Cables, USB (Versions, Specifications, Common Uses, Connectors and Compatibility). Computer Cable Management: Importance of Cable Management for Airflow and Aesthetics, Techniques for Organizing and Securing Cables, Computer Case Management: Types, Form Factors, Importance of Airflow and Ventilation in Computer Cases, Reducing Cable Clutter Inside the Computer Case. Drivers: Definition, Role of Device Drivers, Importance of Drivers for Hardware Functionality, Find and Install Device Drivers. Types of Drivers (Device Drivers, LAN Drivers, Sound Drivers, Graphics Drivers). Cabinet: Role of Cabinet, Types of Computer Cabinets.

UNIT-IV

BIOS: Definition, Importance, Functions, Responsibilities, Role of BIOS in the Boot Process, Importance of BIOS Updates, Procedure for Updating BIOS, Risks and Precautions during BIOS Updates, Common BIOS Error Messages. Antivirus Software: Definition, Features, Significance, Role of Antivirus in Security, Installing and Configuring any Third-Party Antivirus for Optimal Protection. Firewall: Definition, Purpose of Firewalls, Importance in Network Security, Hardware Vs. Software Firewalls, Types of Firewalls (Packet- Filtering Firewalls, Stateful Inspection Firewalls, Proxy Firewalls and Application-Layer Filtering), Limitations and Challenges in Firewall Protection, Real World Examples of Successful Firewall Applications.

12. Suggested Resources for Reading

1. The Indispensable PC Hardware Book (4th Edition), Hans-Peter Messmer, Addison Wesley.
2. PC Hardware: A Beginner's Guide by Ron Gilster, McGraw-Hill Education
3. PC Hardware: The Complete Reference, Craig Zacker., TMH Publication.
4. Mastering PC Hardware & Network, Dr. Ajit Mittal, Dr. Ajay Rana, Khanna Book Publishing Company
5. How Computers Work, Ron White and Timothy Edward Downs, QUE.
6. Fundamentals of Computers, V. Rajaraman, PHI Publication.

Note: Latest edition of text books may be used.

BHN-102P (Practical Based on BHN-102)

Note: Paper will be set at the time of examination. Due weight-age may be given to the Practical note-book and Assignments in evaluation.

1. Examiner will set total of four questions covering the whole syllabus.
2. Student will attempt any three questions.
3. Each question carries 5 marks.
4. Practical file carries 5 marks and 20 marks for practical work and viva-voce.

	COURSE TITLE	PC CONFIGURATION
1.	**Course Code**	BHN-103
2.	**Typology of Course**	Core Discipline
3.	**Course Duration**	120 hours of practicals Practical credits=4 (1 credit = 30 hours of practical based on 8 hrs./week) (1Semester=15weeks) 2-groups will be created for the purpose of Practical for each unit 2-groups will be created for the purpose of Practical for each unit
4.	**Credits**	4 Credits
5.	**Max. Marks and Time**	100Marks (80PR+20IA) i) The exam of this course will be of 3hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
6.	**Job Role**	Field Technician- Computing & Peripherals
7.	**Course Objectives**	The course aims to provide a comprehensive understanding of computer systems, covering fundamental hardware components. It develops practical skills in connecting peripherals, analyze and solve hardware issues logically, and assemble, configure, troubleshoot, and maintain functional computer systems. Additionally, the course focuses on installing Windows, managing drivers, and implementing antivirus software for system security.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to CO1. Define and explain the fundamental components of a computer system, demonstrating a holistic understanding of hardware architecture. CO2. Demonstrate practical skills in connecting and configuring peripherals, including mouse, keyboard, display devices, scanning devices, and hard disk. CO3. Develop critical thinking skills to analyze and solve hardware-related problems, applying a logical and systematic approach to diagnose and address issues. CO4. Assemble, configure, troubleshoot, and maintain a functional computer system, integrating knowledge from all units for practical application. CO5. Successfully install the Windows operating system, manage the installation and updating of drivers, install and update anti-virus software.
9.	**Pedagogy of the Course Work**	Laboratory, Interactive sessions, PPT, Practical and Group Discussions, etc.
10.	**Instructions for paper setters and candidates**	i) Examiner will set total of four questions covering the whole syllabus. ii) Student will attempt any three questions. iii) Each question carries 10 marks. iv) Practical file carries 10 marks and 40 marks for practical work and viva-voce.

11. Course Content

UNIT-I

Introduction To Computer Hardware:

- Overview of Computer Components
- Basic Understanding of CPU, RAM, Storage Devices, Various Ports, Slots (Add On Devices) and Input/Output Devices (Keyboard, Mouse, Monitor, Printer).
- Hands-On Practice Connecting and Configuring Peripherals (Mouse, Keyboard, Display Devices, Scanning Devices, Hard Disk).
- Studying Various Components of The Motherboard.
- Introduction to Hardware Tools and Safety Precautions

UNIT-II

Computer Cables:

- Various Types of Cables Used in Computer, Connectors and Ports used.
- Connecting Cables from SMPS To Motherboard & Hard Disk.
- Establishing Data Connection for Mother Board, Hard Disk, and Drivers.
- Fixing Wires for Power Restart, Fixing Wires for HDD, LED, External USB & Audio Connections.
- Cable Management Techniques.

UNIT-III

System Diagnostics, Troubleshooting and Maintenance:

- Overview of the Troubleshooting Process and the Importance of Systematic Approach
- Introduction To Hardware Diagnostic Tools (PC Toolkit)
- Troubleshooting Common Hardware Problems in Keyboard, Mouse, Mouse, Printer, SMPS, Hard Disk, Motherboard, and Processor.
- Maintenance of Keyboard, Mouse, Mouse, Printer, SMPS, Hard Disk, Motherboard, and Processor.
- Breakdown and Breakthrough of various Hardware Issues.
- Safety Precautions During Troubleshooting.

UNIT-IV

Window Installation:

- Installing Drivers.
- Installing and Updating Anti-Virus Software.
- Hard Disk Partitioning (Creating, formatting & Deleting Partitions) Using DISKPART in Command Prompt.
- Hard Disk Partitioning (Creating, formatting & Deleting Partitions) using Graphical User Interface.

12. Suggested Resources for Reading

1. Comdex Hardware and Networking Course Kit, Gupta, V., Dreamtech Press.
2. IBM-PC and Clones – Hardware Troubleshooting and Maintenance, Govindarajalu, B., Tata McGraw Hill.
3. Mastering PC Hardware & Network, Dr. Ajit Mittal, Dr. Ajay Rana, Khanna Book Publishing Company.
4. PC Hardware: A Beginner's Guide by Ron Gilster, McGraw-Hill Education.
5. PC Hardware: The Complete Reference, Craig Zacker., TMH Publication.

Note: Latest edition of text books may be used.

Semester II

	Course Title	Data Communication and Networks
1.	**Course Code**	BHN-201
2.	**Typology of Course**	Core Discipline
3.	**Course Duration**	60 hours of lectures for Theory Theory credits =4 (1 credit= 15 hours of theory learning based on 4hrs./week) (1 Semester=15 weeks)
4.	**Credits**	4 Credits
5.	**Max. Marks and Time**	100Marks (80TH+20IA) i) The exam of this course will be of 3hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
6.	**Job Role**	Field Technician- Computing & Peripherals
7.	**Course Objectives**	This course aims to familiarize students with diverse network types, detailing their roles in computer environments. It provides a foundational understanding of network security concepts, emphasizing the importance of safeguarding infrastructure. Students will also learn switching techniques, IP address fundamentals and configuration.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1.Develop a comprehensive understanding of computer networks, analyze, and compare various network topologies and understand different network models. CO2.Identify and describe different types of servers, understanding their roles and functionalities in network environments. CO3.Understand transmission media, various networking protocols and the functionalities of network devices. CO4.Develop a basic understanding of network security concepts, recognizing the importance of safeguarding network infrastructure. CO5.Understanding IP addresses and its configuration.
9.	**Pedagogy of the Course Work**	Classroom Lecture, Interactive sessions, PPT, Group Discussions etc.
10.	**Instructions for paper setters and candidates**	i) The syllabus of this paper has been divided into FOUR units. ii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. iii) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. iv) All questions carry equal marks.

11. Course Content

UNIT-I

Computer Network: Introduction, Characteristics, Uses of Computer Networks, Types of Networks Based on Area Coverage: LAN, WAN, MAN, Advantages and Disadvantages of Networks, Network Topologies: Bus Topology, Ring Topology, Star Topology, Tree Topology, Mesh Topology and Hybrid Topology. Network Models: Peer-To-Peer, Protocol, Client/Server Network and Hybrid Type Network. OSI Reference Model, TCP/IP Reference Model, Comparison of OSI with TCP/IP Model. Servers: Different Types of Servers.

UNIT - II

Data Communications: Analog Signal, Digital Signal, Analog Vs Digital Communication, Band Width Limitation, Data Rate of a Channel, Physical Layer: Transmission Media: Guided (Twisted-Pair, Coaxial and Optical Fiber) and Unguided (Radio, Microwave, and Infrared), Switching: Circuit Switching, Packet Switching, Message Switching. Telephone System, Modems. Modulation Techniques: Amplitude Modulation, Frequency Modulation, Phase Modulation. Multiplexing: Frequency Division, Time Division, Wave Length Division.

UNIT-III

Networking Protocols: Introduction, Domain Name System (DNS), DNS Name Space, DNS Servers, FTP, Telnet, DHCP, HTTP, TFTP, SSH, SFTP, SMTP, POP, IMAP, NTP. Network Devices: HUB, Switch, Repeater, Bridges, Routers, Gateway. IEEE Standards: IEEE Standards for LAN and WAN: IEEE Standard 802.3 and Ethernet, IEEE Standard 802.4 Token Bus, IEEE Standard 802.5 Token Ring, IEEE Standard 802.6 Distributed Queue Dual Bus, IEEE Standard 802.2 Logical Link Control. Introduction to Network Security.

UNIT-IV

IP Address: Introduction, IP Address Format, Overview of IP Address Classes: A, B, C, D, and E, Characteristics and Range of Each IP Address Class, Use Cases for Different IP Address Classes. Subnetting- Sub Network, Subnet Mask, Configuration of IP Address in PC, Differentiation Between IPv4 and IPv6. Access Points, Network Interface Card (NIC): Working Principle of NIC, MAC Address. User Management: Overview of User Account, Managing User Account and Password Security: Adding User, Setting and Changing Password, Changing Picture of User, Deleting the User, Setting Account Restriction for The User, Renaming a Local Account.

12. Suggested Resources for Reading

1. Computer Networks, Tanenbaum, Andrew, Fifth Edition, PHI.
2. Data Communication and Networking, Behrouz A. Forouzan, Fourth Edition, McGraw Hill Education.
3. Computer Today, S.K. Basandra, First Edition, Galgotia.
4. Data Communication System, Black, Ulysse, Third Edition, PHI.
5. Data and Computer Communications, Stalling, Ninth Edition, PHI.

Note: Latest edition of text books may be used.

	Course Title	Computing Software
1.	**Course Code**	BHN-202
2.	**Typology of Course**	Core Discipline
3.	**Course Duration**	30 hours of lectures for Theory + 60 hours of practicals Theory credits =2 and Practical credits=2 (1 credit = 15 hours of theory learning based on 2 hrs./week) + (1 credit = 30 hours of practical based on 4 hrs./week) (1 Semester = 15 weeks) 2-groups will be created for the purpose of Practical for each unit
4.	**Credits**	4 Credits
5.	**Max. Marks and Time**	100Marks ((40TH+10IA) +(40PR+10IA)) i) The exam of this course will be of 3hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
6.	**Job Role**	Field Technician- Computing & Peripherals
7.	**Course Objectives**	This course covers essential skills in word processing, spreadsheet management, presentation creation, web browsing, and email communication. Students will learn document handling, formatting, and advanced features in Microsoft office applications. Additionally, they will acquire proficiency in web browsing, email etiquette, organization strategies, and customization options, enhancing their overall digital literacy.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1. Develop proficiency in using office suite applications, including word processing, spreadsheets, and presentation tools, for efficient document creation, data analysis, and presentation development. CO2. Develop technological literacy by understanding and utilizing the latest features and tools available in office suite applications. CO3. Students will be able to develop engaging presentations, mastering slide design, animations, and transitions, enabling effective communication through multimedia slideshows. CO4. Understand basic communication tools, web surfing techniques, e-governance website usage, and understand email security settings, two- factor authentication, and safe practices for online communication.
9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions etc.
10.	**Instructions for paper setters and candidates**	i) The syllabus of this paper has been divided into FOUR units. ii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. iii) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. iv) All questions carry equal marks.

11. Course Content

UNIT-1

Word Processing Package: Opening, Saving and Closing an Existing Document, Renaming and Deleting Files, Using Styles and Templates: Introduction to Templates and Styles, Applying, Modifying and Creating New(Custom) Styles, using a Template to Create a Document, Creating a Template, Editing a Template, Organizing Templates, Examples of Style Use, Changing Document Views, Moving Quickly Through a Document.

Working With Text: Select, Cut, Copy, Paste, Find and Replace, Inserting Special Characters, Setting Tab Stops and Indents, Checking Spelling and Grammar, Autocorrect, Using Built-In Language Tools, Word Completion, Autotext, Formatting Text: Using Styles, Formatting Paragraphs, Formatting Characters, Autoformatting, Creating Lists.

Formatting Pages: Using Layout Methods, Creating Headers and Footers, Numbering Pages, Changing Page Margins, Adding Comments to a Document, creating a Table of Contents, Creating Indexes and Bibliographies, Printing a Document, Using Mail Merge, Tracking Changes to a Document, Using Fields, Linking to Another Part of a Document, Using Master Documents, Creating Fill-In Forms.

UNIT-II

Spreadsheet Package: Worksheet Basics, Data Entry in Cells: Entry of Numbers, Text and Formulae, Moving Data in a Worksheet, Moving Around in a Worksheet, Selecting Data Range, Using the Interface (Toolbars, Menus), Editing Basics, Working with Workbooks, Cell Referencing. Formatting and Calculations: Using Auto Fill, Working with Formulae, Efficient Data Display with Data Formatting (Number Formatting, Date Formatting Etc.), Working with Ranges, Worksheet Printing.

Working with Graphs and Charts: Creating Embedded Chart Using Char Wizard, Sizing and Moving Parts, Updating Charts, Changing Chart Types, Chart Wizard, Adding Titles, Legends and Gridlines, Printing Charts, Database Management. Finding Records with Data Form, Adding/Deleting Records, Filtering Records in a Worksheet. Functions and Macros: Worksheet Creating Macros, Recording Macros, Running Macros, Assigning Macros to Buttons, Defining Macros from Scratch. Multiple Worksheets.

UNIT-III

Presentation Package: Introducing Power Point, Use of Power Point Presentation, Creating New Slides Saving, Opening, and Printing. Home Tab: New Slide, Layout, Reset, Delete, Setting Text Direction, Align Text, Convert to Smart Art, Drawing Options. Insert Tab: Table, Picture, Clipart, Photo Album, Smart Art, Shapes and Chart, Movie and Sound, Hyperlink and Action, Text Box, Word Art, Object.

Design Tab: Page Setup Options, Slide Orientation, Applying Various Themes, Selecting Background Style, and Formatting it. Animations Tab: Custom Animation for Entrance, Exit and Emphasis, Applying Slide Transition, Setting Transition Speed and Sound, Animation on Rehears Timing. Slide Show &View Tab: Start Slide Show Options, Setup Options. View Tab: Presentation Views, Colours, and Window Option.

UNIT-IV

Web Browsing: Popular Web Browsing Softwares, Search Engines, Search for Content Accessing Web Browser, Using Favorites Folder, Downloading Web Pages, Printing Web Pages, Understanding URL, Surfing the Web, Using E-Governance Website.

Communication: Overview of E-Mail and its Evolution, Creating and Customizing E-Mail Account, Understanding E-Mail User Interface and Layout, Basic Email Etiquette, Essential Email Features, Composing, Sending, and Receiving Emails, Attachments and File Sharing, Labels and Filters for Organizing Emails, Advanced Search Techniques.

Organization Strategies: Inbox Organization and Management, Priority Inbox and Tabbed Inbox Features, Using Folders (Labels) Effectively, Archiving and Deleting Emails, Advanced Settings, and Customization: Customizing E-Mail Settings for Personal Preferences, Signature Creation and Customization, Vacation Responders and Out-of-office Settings, Security Settings and Two-Factor Authentication, Mobile and offline Access: Configuring E-Mail on Mobile Devices, offline Access, and Synchronization.

12. Suggested Resources for Reading

1. Microsoft Windows Networking Essentials 1st Edition, Darril Gibson, Sybex.
2. Computer Fundamentals, Sinha P.K. & Sinha Priti, BPB Publications.
3. DOS 6.2 Quick Reference, Mathur Rajiv, Galgotia.
4. Introduction to Computers with MS - Office, Leon, TMH Publication
5. Using the Internet, Kasser, Barbara, PHI, 4th ed., New Delhi.
6. Using the World Wide Web, Wall, David A. & Others, PHI, 2nd ed., New Delhi.

Note: Latest edition of text books may be used.

BHN-202P (Practical Based on BHN-202)

Note: Paper will be set at the time of examination. Due weight-age may be given to the Practical note-book and Assignments in evaluation.

1. Examiner will set total of four questions covering the whole syllabus.
2. Student will attempt any three questions.
3. Each question carries 5 marks.
4. Practical file carries 5 marks and 20 marks for practical work and viva-voce.

	Course Title	Network Configuration
1.	**Course Code**	BHN-203
2.	**Typology of Course**	Core Discipline
3.	**Course Duration**	120 hours of practicals Practical credits=4 (1 credit = 30 hours of practical based on 8 hrs./week) (1Semester=15weeks) 2-groups will be created for the purpose of Practical for each unit 2-groups will be created for the purpose of Practical for each unit
4.	**Credits**	4 Credits
5.	**Max. Marks and Time**	100Marks (80PR+20IA) i) The exam of this course will be of 3hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
6.	**Job Role**	Field Technician- Computing & Peripherals
7.	**Course Objectives**	This course aims to provide a comprehensive understanding of networking tools, connectors, cables, and testing instruments. Participants will learn to establish LAN connections, configure internet connectivity, and implement Wi-Fi setups. The course emphasizes hands-on skills in cable termination, network testing, and troubleshooting.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to CO1. Proficiently utilize networking tools like crimping and punching tools. CO2. Demonstrate expertise in working with various connectors, including RJ-45 and Fiber LC. CO3. Construct and troubleshoot LAN connections using UTP cables, switches, and testing tools**.** CO4. Execute networking commands for configuration and diagnostics. CO5. Configure and secure internet and Wi-Fi connectivity, demonstrating knowledge of different internet connection types.
9.	**Pedagogy of the Course Work**	Laboratory, Interactive sessions, PPT, Practical and Group Discussions, etc.
10.	**Instructions for paper setters and candidates**	i) Examiner will set total of four questions covering the whole syllabus. ii) Student will attempt any three questions. iii) Each question carries 10 marks. iv) Practical file carries 10 marks and 40 marks for practical work and viva-voce.

11. Course Contents

UNIT-I

Working Using Various Networking Tools:

Crimping Tool, Punching Tool, Punch Down Tool, Multi Punch Tool, RJ45 Punch Down Tool, Computer Punch Tool

Cable Cutter, I/O Device, I/O Face Plate, I/O Box, Lan Cards

Working Using Various Connectors:

RJ-11, RJ-45, F-Type, ST (Straight Tip)

SC (Subscriber Connector or Standard Connector)

Fiber LC (Local Connector)

MT- RJ (Mechanical Transfer Registered Jack)

USB (Universal Serial Bus)

UNIT-II

Networking Cables, Patch Cords and Testing Tools:

Working of Different Cables: Coaxial Cable, Unshielded Twisted Pair (UTP), Types of UTP Cable, Shielded Twisted Pair Cable, Fiber Optics.

Patch Cords: UTP Patch Cords, Fiber Patch Cords. Cable Manager, Cable Tag, Cable Stripper, Scissors.

Testing Tools: LAN Testers, Application Tools- IP Scanner, Test Network Speed, Bandwidth Monitor 2 Lite Edition, Netbrute Scanner.

UNIT-III

Establishing Lan Connection:

Patching of LAN Cable, Strip Cable End, Untwist Wire Ends.

Arrange Wires, Trim Wires to Size, Attach Connector, Check, Crimp, Test.

Setting Up of LAN Using UTP Cables, Switch, LAN Tester, Switch Rack, Patch Panel, Cable Manager, Cable Tagging, I/O Box, I/O Face Plate, Patch Cords, Connectors, Terminals.

Networking Commands:

ping, ipconfig, fstat, iproute, route, netstat, ipconfig/release, ipconfig/renew.

Unit-IV

Internet Connectivity:

Different Types of Internet Connection (Wireless, Mobile, Hotspot, Dial-Up, Broadband, DSL, Cable, Satellite, ISDN.)

Wi-Fi Connectivity:

Basic Configuration of Wi-Fi Device or Access Point, IP Address Configuration, DNS Address, Security and Password, Admin Login and Password, MAC Filtering and Monitoring.

12. Suggested Resources for Reading

1. CCNA Cisco Certified Network Associate Study Guide 3rd Edition, McGraw Hill Education.
2. Cisco Networking Essentials, Troy Macmillan, Sybex.
3. Comdex Hardware and Networking Course Kit, Vikas Gupta, 1st Edition, Dreamtech Press.
4. Networking: The Complete Reference 1st Edition, McGraw Hill Education.
5. Wireless Communication: Principle and Practices, Theodore S. Rappaport, Pearson Publication.

Note: Latest edition of text books may be used.